Evaluation & Quality Systems

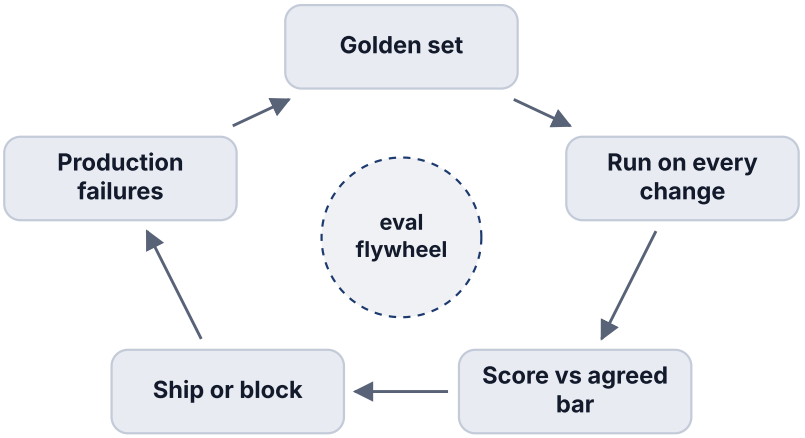
1. Why evals are the centre of gravity

Evaluation is the **control system** of AI engineering. Without it you can't choose a model, approve a prompt change, or answer the only question executives ask: 'is it good, and is it getting better?'. Amateurs build and eyeball; professionals define what good means, encode it as a suite, and let the suite gate every change. In a statement of work, an eval threshold turns 'is it done?' from an argument into a measurement.

2. The golden set

- **30 to 100 real, hand-labelled examples** to start: real inputs with expected outcomes
- Deliberately mix easy cases, hard cases and known failure modes
- **The flywheel**: every bad production output becomes a permanent test case, so the same failure can never return unnoticed

✎ DIAGRAM



Failures from production feed back into the golden set, so the suite only grows.

3. Scoring

- **Exact checks** wherever possible: schemas validate, SQL returns the right result, code passes tests. Free and unarguable
- **LLM-as-judge** for open-ended qualities (helpfulness, faithfulness, tone): a strong model grades against a written rubric
- Judge biases: **position** (favours the first answer: swap and average), **verbosity** (rewards length: tell it not to), **self-preference** (favours its own family: use another provider), **rubric drift** (vague rubrics score vibes)

- **Calibrate** the judge against human labels and only trust it once agreement is high

4. Gating changes

- Run the suite in **CI on every change**: prompt, model, chunking, documents, settings
- Agree the **pass bar** up front, so shipping is a rule, not a judgment call
- Below the bar → the change is blocked, not rolled back after it's live
- After launch: **online evals** on a sample of live traffic, plus thumbs-down, escalations and drift monitoring

5. Metrics by layer

- **Retrieval**: recall@K, precision@K, MRR, nDCG
- **RAG (RAGAS family)**: **context precision** (was the context relevant?), **context recall** (did it contain everything needed?), **faithfulness** (claims backed by context?), **answer relevance** (does it answer the question?)
- **Generation**: exact match / F1 for short facts; **BLEU / ROUGE** are word-overlap scores and weak for open-ended answers; **pointwise** (score one answer) vs **pairwise** (A vs B, giving a win rate)
- **Classification**: precision, recall, F1, confusion matrix
- **Business**: **containment** (resolved without a human), escalation rate, resolution rate, CSAT, **cost per resolved case**

✎ DIAGRAM



Name a metric for each layer; 'we'd run evals' alone isn't an answer.

6. Rigour

- Enough examples to trust a difference (2 points on 30 questions may be noise)
- Outputs vary, so run several times and average
- Keep a held-out test set you never tune against

SAY IT

"I'd build a golden set from real questions, measure retrieval and answers separately, gate every change in CI against an agreed bar, and monitor live traffic after launch."